



GROUP LAB PROJECT

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Course: **Programming Fundamental**

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Topic: Car Workshop Management

Objective:

The objective of this lab project is to create a program for Auto Workshop, which enables customers to select their preferred brand, model, services, and parts for their vehicles, and then calculates the total bill including any discounts and handles payments.

Code Analysis:

1. Header Files and Namespace:

The program begins by including necessary header files such as '`<iostream>`', '`<string>`' and '`<cstdlib>`'. The 'namespace std' is also used. This library ('`<cstdlib>`') is used for deleting previous output.

2. Function Definitions:

1. 'displayOptions': This function displays a list of options based on the provided array of strings. It takes an array of strings, the size of the array, and a name (for display purposes) as parameters, and returns the index of the selected option.

2. 'calculateCost': This function calculates the cost based on the selected car, service, and part.

3. Main Function:

Initialization: Variables for brand, model, and cost are initialized. Arrays for model names, brand names, choices, parts, and their corresponding values/prices are defined. Conditional statements are used for user choices for modification or adding in services. Loop are used for repeating of statements.

Brand and Model Selection: Users are prompted to select a brand and then a model from the available options.

Service Selection: Users can select various services they require for their car.

Shopping Cart: The selected items are added to a shopping cart, and the total cost is updated accordingly.

Editing Cart: Users have the option to edit their shopping cart.

Total Bill Calculation: After finalizing the selections, the program calculates the total bill, including any discounts if applicable.

Payment Handling: Users are prompted to input their payment, and change is returned.

Experimental Results:

Upon executing the program, the user is greeted with the available brands and prompted to make selections for brand, model, services, and parts. After confirming their choices, the program calculates the total bill and handles payments.

Working of the Program:

1. Initialization:

The program starts by displaying the header "HAFIZ AUTO WORKSHOP" and initializes necessary variables and arrays for brand names, model names, services, and their corresponding values/prices.

2. Brand and Model Selection:

The user is prompted to select a brand from the available options (Honda, Toyota, Hyundai, and Suzuki). Based on the selected brand, the user is then prompted to choose a model from the available options for that brand.

3. Service Selection:

After selecting the model, the user is presented with a list of available services (Denting, Painting, Polishing, Washing, and Waxing). After selecting the available services, the user is presented with a list of car body parts (Roof, Bonnet, Front Bumper, Rear Bumper, Left Fender, Right Fender, Left Quarter Panel, Right Quarter Panel, Front

Door Left, Front Door Right, Rear Door Left, Rear Door Right, Full Body, and Full/Wash). The user can choose one or more services they require.

4. Shopping Cart:

The selected items (brand, model, services) are added to a shopping cart, and the total cost is calculated based on the selected car model, service, and parts. The program displays the current bill with the selected options.

5. Editing Cart:

Users have the option to edit their shopping cart by modifying their selections for services and parts. The program recalculates the total cost based on the updated selections.

6. Total Bill Calculation:

After finalizing the selections, the program calculates the total bill, including any discounts if applicable.

7. Payment Handling:

The user is prompted to input the payment amount. The program calculates the change to be returned to the user after deducting the payment from the total bill.

8. Conclusion:

The program provides the user with a detailed breakdown of their selections and the corresponding costs. Upon completion of the transaction, the user receives the total bill amount and any change due.

Program Code:

```
#include <iostream>

#include <string>

#include <cstdlib> // For system("cls")

using namespace std;

int displayOptions(string* opt, int size, string name) {
```

```

for(int i = 0; i < size; i++) {
    cout << i + 1 << ". " << opt[i] << endl;
}
int input;
cout << "\nChoose your " << name << " option #: ";
cin >> input;
input--;
system("cls");
return input;
}

double calculateCost(double car, double choice, int part)
{
    double total = car * choice * part;
    return total;
}

int main() {
    cout << "\t\t===== HAFIZ AUTO WORKSHOP ===== \n\n" << endl;

    char ch;
    double cost = 0;
    double modelCost[20];
    string modelName[20];
    string brandName;
    int choice;
    string brands[] = {"Honda", "Toyota", "Hyundai", "Suzuki"};

    string honda[] = {"Civic", "City", "BR-V", "CR-V"};

```

```
double carsValue[] = {1.2, 1.5, 1.9, 1.1};
```

```
string toyota[] = {"Fortuner", "Grande", "Vitz", "Prius"};
```

```
double cars1Value[] = {2.5, 1.7, 1.2, 1.7};
```

```
string hyundai[] = {"Tuscon", "Elantra", "Sonata", "Picanto"};
```

```
double cars2Value[] = {1.7, 1.5, 1.5, 1.1};
```

```
string suzuki[] = {"Cultus", "Swift", "Alto", "Ciaz"};
```

```
double cars3Value[] = {0.7, 0.5, 0.3, 0.9};
```

```
string choices[] = {"Denting", "Painting", "Polishing", "Washing", "Wax"};
```

```
double choiceValue[] = {2.3, 2.0, 1.3, 1.0, 1.5};
```

```
string parts[] = {"Roof", "Bonnet", "Front Bumper", "Rear Bumper", "Left Fender", "Right Fender", "Left Quarter Panel", "Right Quarter Panel", "Front Door Left", "Front Door Right", "Rear Door Left", "Rear Door Right", "Full Body", "Full/Wash"};
```

```
int prices[] = {8000, 6000, 2500, 2500, 3500, 3500, 3600, 3600, 4500, 4500, 4900, 4900, 50000, 500};
```

```
int cartCars[20];
```

```
int cartChoice[20];
```

```
int cartPart[20];
```

```
double cartPrice[20];
```

```
int index = 0;
```

```
int model;
```

```
cout << "\t\t===== AVAILABLE BRANDS ===== \n\n" << endl;
```

```
cout << "1. Honda\n";
```

```
cout << "2. Toyota\n";
```

```

cout << "3. Hyundai\n";
cout << "4. Suzuki\n";
cout << "\nSelect your brand #: ";
cin >> choice;
system("cls");
cout << "\n\n\t\t===== AVAILABLE MODELS ===== \n\n" << endl;
switch (choice) {
    case 1:
        model = displayOptions(honda, 4, "Model");
        for (int i = 0; i < 4; ++i) {
            modelCost[i] = carsValue[i];
            modelName[i] = honda[i];
        }
        brandName = "Honda";
        break;
    case 2:
        model = displayOptions(toyota, 4, "Model");
        for (int i = 0; i < 4; ++i) {
            modelCost[i] = cars1Value[i];
            modelName[i] = toyota[i];
        }
        brandName = "Toyota";
        break;
    case 3:
        model = displayOptions(hyundai, 4, "Model");
        for (int i = 0; i < 4; ++i) {
            modelCost[i] = cars2Value[i];
            modelName[i] = hyundai[i];

```

```

    }
    brandName = "Hyundai";
    break;
case 4:
    model = displayOptions(suzuki, 4, "Model");
    for (int i = 0; i < 4; ++i) {
        modelCost[i] = cars3Value[i];
        modelName[i] = suzuki[i];
    }
    brandName = "Suzuki";
    break;
default:
    cout << "Invalid choice. Please try again.\n";
    // return 0;
}

cout << "\n\n\t\t===== AVAILABLE SERVICES ===== \n\n" << endl;

while(true) {
    int choice = displayOptions(choices, 5, "Choice");
    cout << "\n\n\t\t===== OPTION of CAR BODY ===== \n\n" << endl;
    int part = displayOptions(parts, 14, "Part");
    cout << "\n\n\t\t===== CURRENT BILL ===== \n\n" << endl;

    cartCars[index] = model;
    cartChoice[index] = choice;
    cartPart[index] = part;
}

```



```
cartPrice[index] = calculateCost(modelCost[model], choiceValue[choice], prices[part]);  
cost += cartPrice[index];
```

```
for (int i = 0; i <= index; i++) {  
    cout << brandName << ", " << modelName[cartCars[i]] << ", " << parts[cartPart[i]] << ",  
" << choices[cartChoice[i]] << ": Rs." << cartPrice[i] << endl;  
}
```

```
cout << "\nDo you want to change in any sevices? (y/n): ";  
cin >> ch;
```

```
if (ch == 'y' || ch == 'Y')  
    { int editIndex=1;  
      editIndex--;
```

```
        choice = displayOptions(choices, 5, "Choice");  
        part = displayOptions(parts, 14, "Part");  
        cartChoice[editIndex] = choice;  
        cartPart[editIndex] = part;  
        cartPrice[editIndex] = calculateCost(modelCost[model], choiceValue[choice],  
prices[part]);  
        cost = 0;  
        for (int i = 0; i <= index; i++)  
            { cost += cartPrice[i];  
              }  
        }  
    }  
    for (int i = 0; i <= index; i++) {
```

```

        cout << brandName << ", " << modelName[cartCars[i]] << ", " << parts[cartPart[i]] << ",
" << choices[cartChoice[i]] << ": Rs." << cartPrice[i] << endl;
    }

    // cout<<"\nYour Total Cost is: Rs. " << cost << endl;

    cout << "\nDo you want more services? (y/n): ";

    cin >> ch;

    if(ch!='y' && ch!='Y')
        break;

    system("cls");

    index++;
}

// while (ch == 'y' || ch == 'Y');

cout << "\n\n\t\t===== YOUR TOTAL BILL ===== \n\n" << endl;

for (int i = 0; i <= index; i++) {
    cout << brandName << ", " << modelName[cartCars[i]] << ", " << parts[cartPart[i]] << ",
" << choices[cartChoice[i]] << ": Rs." << cartPrice[i] << endl;
}

cout << "\nTotal Cost is: Rs. " << cost << endl;

double discount;

double diis;

double disc;

char dis;

cout<<"\nDo you want to add discount?(y/n) ";

cin>>dis;

if(dis == 'y' || dis ==
'Y'){ cout<<"\nEnter disount %: ";

```

```

        cin>>discount;

        system("cls");

        diis=(discount/100)*cost;

        disc=cost-diis;

        cout << "\n\n\t\t===== YOUR TOTAL BILL ===== \n\n" << endl;

        for (int i = 0; i <= index; i++) {

            cout << brandName << ", " << modelName[cartCars[i]] << ", " << parts[cartPart[i]] << ",
" << choices[cartChoice[i]] << ": Rs." << cartPrice[i] << endl;

        }

        cout << "\nTotal Price is: Rs. " << cost << endl;

        cout << "\nTotal Payable Amount After Discount is: Rs. " << disc << endl;

    }

    else{    cout << "\n\n\t\t===== YOUR PAYABLE BILL ===== \n\n" << endl;

        for (int i = 0; i <= index; i++) {

            cout << brandName << ", " << modelName[cartCars[i]] << ", " << parts[cartPart[i]] << ",
" << choices[cartChoice[i]] << ": Rs." << cartPrice[i] << endl;

        }

        cout << "Payable Amount is:          Rs. " << cost << endl;

    }

    double payment=0.0;

    double change=0.0;

    do{

        cout << "Payment Received:          Rs. ";

        cin>>payment;

        if(payment>=cost){ chan

            ge=payment-cost;

            cout<<"Balance Returned to customer: Rs. " << change;

```

```
}  
else  
    if(payment>=disc){ c  
        hange=payment-disc;  
        cout<<"Balance Returned to customer: Rs. "<<change;  
        break;  
    }  
    else{  
        cout<<"Insuffient Amount....";  
    }  
}while(payment<cost);  
  
return 0;  
}
```

Screenshot:

Starting:

1st Selections:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"

===== HAFIZ AUTO WORKSHOP =====

===== AVAILABLE BRANDS =====

1. Honda
2. Toyota
3. Hyundai
4. Suzuki

Select your brand #:
```

2nd Selections:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"

===== AVAILABLE MODELS =====

1. Tuscon
2. Elantra
3. Sonata
4. Picanto

Choose your Model option #:
```

3rd Selections:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"

===== AVAILABLE SERVICES =====

1. Denting
2. Painting
3. Polishing
4. Washing
5. Wax

Choose your Choice option #: _
```

4th Selections:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"

===== OPTION of CAR BODY =====

1. Roof
2. Bonnet
3. Front Bumper
4. Rear Bumper
5. Left Fender
6. Right Fender
7. Left Quarter Panel
8. Right Quarter Panel
9. Front Door Left
10. Front Door Right
11. Rear Door Left
12. Rear Door Right
13. Full Body
14. Full/Wash

Choose your Part option #: _
```

Current Bill:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"

===== CURRENT BILL =====

Hyundai, Tuscon, Full Body, Washing: Rs.85000

Do you want to change in any services? (y/n):
```

Modification stage 1:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"

===== CURRENT BILL =====

Hyundai, Tuscon, Full Body, Washing: Rs.85000

Do you want to change in any services? (y/n): y
1. Denting
2. Painting
3. Polishing
4. Washing
5. Wax

Choose your Choice option #: _
```

Modification stage 2:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"

1. Roof
2. Bonnet
3. Front Bumper
4. Rear Bumper
5. Left Fender
6. Right Fender
7. Left Quarter Panel
8. Right Quarter Panel
9. Front Door Left
10. Front Door Right
11. Rear Door Left
12. Rear Door Right
13. Full Body
14. Full/Wash

Choose your Part option #: _
```

After Modification Bill:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"  
Hyundai, Tuscon, Full Body, Wax: Rs.127500  
Do you want more services? (y/n): _
```

Adding more Services:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"  
1. Denting  
2. Painting  
3. Polishing  
4. Washing  
5. Wax  
  
Choose your Choice option #: 4_
```

Again Services Options:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"  
  
===== OPTION of CAR BODY =====  
  
1. Roof  
2. Bonnet  
3. Front Bumper  
4. Rear Bumper  
5. Left Fender  
6. Right Fender  
7. Left Quarter Panel  
8. Right Quarter Panel  
9. Front Door Left  
10. Front Door Right  
11. Rear Door Left  
12. Rear Door Right  
13. Full Body  
14. Full/Wash  
  
Choose your Part option #:
```

Bill after Adding Services:


```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"

===== CURRENT BILL =====

Hyundai, Tuscon,Full Body, Wax: Rs.127500
Hyundai, Tuscon,Full/Wash, Washing: Rs.850

Do you want to change in any sevices? (y/n): _
```

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"

===== CURRENT BILL =====

Hyundai, Tuscon,Full Body, Wax: Rs.127500
Hyundai, Tuscon,Full/Wash, Washing: Rs.850

Do you want to change in any sevices? (y/n): n
Hyundai, Tuscon, Full Body, Wax: Rs.127500
Hyundai, Tuscon, Full/Wash, Washing: Rs.850

Do you want more services? (y/n): n

===== YOUR TOTAL BILL =====

Hyundai, Tuscon, Full Body, Wax: Rs.127500
Hyundai, Tuscon, Full/Wash, Washing: Rs.850

Total Cost is: Rs. 128350

Do you want to add discount?(y/n) _
```

Discount Option:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"

===== CURRENT BILL =====

Hyundai, Tuscon, Full Body, Wax: Rs.127500
Hyundai, Tuscon, Full/Wash, Washing: Rs.850

Do you want to change in any sevices? (y/n): n
Hyundai, Tuscon, Full Body, Wax: Rs.127500
Hyundai, Tuscon, Full/Wash, Washing: Rs.850

Do you want more services? (y/n): n

===== YOUR TOTAL BILL =====

Hyundai, Tuscon, Full Body, Wax: Rs.127500
Hyundai, Tuscon, Full/Wash, Washing: Rs.850

Total Cost is: Rs. 128350

Do you want to add discount?(y/n) y
Enter disount %:
```

Bill after Adding Discount:

```
"D:\PF project 1\Car Work shop\Auto Workshop.exe"

===== YOUR TOTAL BILL =====

Hyundai, Tuscon, Full Body, Wax: Rs.127500
Hyundai, Tuscon, Full/Wash, Washing: Rs.850

Total Price is: Rs. 128350

Total Payable Amount After Discount is: Rs. 115515
Payment Received: Rs.
```

Payment Option:

"D:\PF project 1\Car Work shop\Auto Workshop.exe"

===== YOUR TOTAL BILL =====

Hyundai, Tuscon, Full Body, Wax: Rs.127500

Hyundai, Tuscon, Full/Wash, Washing: Rs.850

Total Price is: Rs. 128350

Total Payable Amount After Discount is: Rs. 115515

Payment Received: Rs. 120000

Balance Returned to customer: Rs. 4485

Process returned 0 (0x0) execution time : 228.761 s

Press any key to continue.